

MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (2 hours) April 1st, 2017

Instructions: Attempt any three out of the following four questions on both pages.
Mention which three you would like to have graded. Open book and open notes.

1. Suppose X is exponentially distributed with density

$$f_X(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & x \leq 0 \end{cases}$$

Consider $Z = h/X$, where $h > 0$.

- (a) Find $f_Z(z)$, the density of $Z = h/X$.
- (b) Given an iid sample $X_1, \dots, X_n$, we have $Z_1 = h/X_1, \dots, Z_n = h/X_n$. Find the MLE (maximum likelihood estimator) of h , denoted by $\hat{h}_{MLE}$.
- (c) Find the asymptotic variance of $\hat{h}_{MLE}$.

2. One coin is randomly selected from a collection of eight. Seven of these coins are scrupulously fair, but the eighth is biased, landing heads with probability 0.7.

- (a) The selected coin is tossed twice, and it lands heads both times. In view of this information, what is the probability that the selected coin was the biased one?

- (b) A different coin is selected from the remaining 7 with 2 heads being seen when tossed twice. What is the probability that the second selected coin was biased?

3. Suppose that Y has a log-normal distribution with density $f(y) = \frac{1}{y\sqrt{2\pi\sigma^2}} e^{-\frac{(\log(y)-\mu)^2}{2\sigma^2}}$, $y > 0$

- (i) Show that $Z = \log(Y)$ has a $Normal(\mu, \sigma^2)$ distribution.
- (ii) Calculate the expected value and variance of Y .
- (iii) Show that if $Y_1, Y_2, \dots, Y_n$ are independent and with the same distribution as Y then $\prod_{i=1}^n Y_i$ also has a log-normal distribution.

4. Suppose X has a $Beta(\theta, 1)$ distribution. Then the most powerful test of level α for testing the null hypothesis $\theta = 1$ vs the alternative hypothesis $\theta = 2$ rejects the null hypothesis when $X > 1 - \alpha$ and otherwise does not reject. Check that this is the UMP test against the alternative hypothesis $\theta > 1$